

HDOC Day-13 Assignment

AIM :

To write a menu-driven C code using a switch-case statement that takes a positive integer as input and performs one of the following operations according to the user's choice .

- * count the number of even digits and odd digits.
- * calculate the sum of even digits and the sum of odd digits separately.

ALGORITHM ;

1. Start
2. Declare variables ; num , temp , choice , digit , *even_count* , *odd_count* , *Even_sum* , and *odd_sum*
3. display a menu with the following choices ;
 - * choice 1 ; count even and odd digits.
 - * choice 2 ; find the sum of even and odd digits.
4. input the user's selected choice.
5. input a positive integer sum
6. use a switch-case statement on choice ;

CASE 1 -

- Initialize *even_count* = 0 and *odd_count* = 0.
- Copy num to temp.
- Loop while temp > 0:
 - Extract the last digit: *digit* = temp % 10.

- If $\text{digit} \% 2 == 0$, increment `even_count` by 1.
- Else, increment `odd_count` by 1.
- Remove the last digit: $\text{temp} = \text{temp} / 10$.
- Print `even_count` and `odd_count`.

CASE 2;

- Initialize `even_sum = 0` and `odd_sum = 0`.
- Copy `num` to `temp`.
- Loop while `temp > 0`:
 - Extract the last digit: $\text{digit} = \text{temp} \% 10$.
 - If $\text{digit} \% 2 == 0$, add `digit` to `even_sum`.
 - Else, add `digit` to `odd_sum`.
 - Remove the last digit: $\text{temp} = \text{temp} / 10$.
- Print `even_sum` and `odd_sum`.
- **Default:**
- Print "Invalid Choice!".

7. stop

Test Case Table -

Test case	Input number	Choice selected by user	Expected output	description
1.	84213	1	Even digits = 3, Odd digits = 2	Test choice 1 with mixed digits
2.	6842	1	Even digits = 4, Odd digits = 0	Test choice 1 with all even digits
3.	7531	2	Sum of even digits = 0, Sum of odd digits = 16	Test choice 2 with all odd digits (7+5+3+1=16)

C PROGRAM CODE ;

```
OW PICU 5.07

#include <stdio.h>

int main() {
    int num, temp, choice, digit;
    int even_count = 0, odd_count = 0;
    int even_sum = 0, odd_sum = 0;

    printf("- MENU -\n");
    printf("1. Count number of even and odd digits\n");
    printf("2. Find sum of even and odd digits\n");
    printf("Enter your choice (1 or 2): ");
    scanf("%d", &choice);

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    switch(choice) {
        case 1:
            temp = num;
            while(temp > 0) {
                digit = temp % 10;
                if(digit % 2 == 0) {
                    even_count++;
                } else {
                    odd_count++;
                }
                temp = temp / 10;
            }
            printf("\nNumber: %d\n", num);
            printf("Even_dig = %d, Odd_dig = %d\n", even_count, odd_count);
            break;

        case 2:
            temp = num;
            while(temp > 0) {
                digit = temp % 10;
                if(digit % 2 == 0) {
                    even_sum = even_sum + digit;
                } else {
                    odd_sum = odd_sum + digit;
                }
                temp = temp / 10;
            }
            printf("\nNumber: %d\n", num);
            printf("Sum_even = %d, Sum_odd = %d\n", even_sum, odd_sum);
            break;

        default:
            printf("\nInvalid Choice! Please enter 1 or 2.\n");
    }

    return 0;
}
```

CODE EXECUTION AND OUTPUT SCREENSHOT ;

```
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % nano 13q1.c
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % gcc 13q1.c -o 13q1
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % ./13q1
- MENU -
1. Count number of even and odd digits
2. Find sum of even and odd digits
Enter your choice (1 or 2): 1
Enter a positive integer: 84213

Number: 84213
Even_dig = 3, Odd_dig = 2
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % ./13q1
- MENU -
1. Count number of even and odd digits
2. Find sum of even and odd digits
Enter your choice (1 or 2): 1
Enter a positive integer: 6842

Number: 6842
Even_dig = 4, Odd_dig = 0
[manalithakar@Unknown_8e:80:9d:9c:62:11 ~ % ./13q1
- MENU -
1. Count number of even and odd digits
2. Find sum of even and odd digits
Enter your choice (1 or 2): 2
Enter a positive integer: 7531

Number: 7531
Sum_even = 0, Sum_odd = 16
manalithakar@Unknown_8e:80:9d:9c:62:11 ~ %
```